

Hoang Nam Dang

Research Engineer, MIT CSAIL

✉ hnd@mit.edu ☎ (515) 694-8353 📍 Cambridge, MA 🌐 6namdang.github.io 🐙 6namdang 📄 dang004
 📄 Google Scholar 🆔 0009-0008-9267-493X

Summary

Research engineer at MIT CSAIL (MANTIS AI) focused on AI for Science, integrating vision-language models, spatial transcriptomics, and interpretable machine learning for biology and healthcare.

Education

B.S. Gannon University, Computer Science Erie, PA
May 2025
 • GPA: 4.0/4.0
 • Awarded J.J. Duratz Research Award (2025)

Research Interests

- AI for Science (AI4S), with emphasis on safe and interpretable models aligned with human goals
- Machine learning and computer vision for biology, additive manufacturing, and security

Research Experience

MIT CSAIL — MANTIS AI, Engineer Remote / Cambridge, MA
Jan 2026 – present
 Supervisor: Prof. Manolis Kellis

- Architecting a vision-language model framework to integrate high-resolution spatial transcriptomics with single-cell modalities for Alzheimer's disease selective vulnerability
- Fine-tuning transformers on large-scale genomic datasets to identify gene-driven resilience (e.g., RORB) with interpretable cellular health mapping across early-stage AD clusters
- Implementing differential expression gene (DEG) analysis as grounded biological embeddings linking transcriptomic signals to clinical outcomes

Gannon University — Network Security Lab, Security Research Assistant Jan 2024 – Jan 2026
 Supervisor: Prof. Ronny Bazan-Antequera

- Proposed a cost-effective, reliable, serverless AI firewall adaptable to on-prem and cloud deployments
- Built and fine-tuned an LSTM on CICIDS2017 (98% accuracy), outperforming SVM, K-NN, and baseline neural networks

Gannon University — 3D Printing Lab, Computer Vision Research Assistant Jan 2025 – May 2025
 Supervisor: Prof. Longfei Zhou

- Detected invisible 3D-printing errors using Bayesian methods and Gaussian distributions
- Proposed VGG-19 segmentation with CNN error detection, achieving ~10× faster computation

Gannon University — Computer Science and Information Lab, AI Research Assistant May 2023 – present
 Supervisors: Prof. Longfei Zhou and Prof. Kefei Wang

- Applied CatBoost, XGBoost, and SVM to predict student matriculation rates
- Studied pre-training and post-training techniques including SFT, GPO, GRPO, DPO, and 4-bit quantization for efficient LLMs

Gannon University — Biology Cell Lab , Biology Research Assistant Supervisors: Prof. Nicholas Conklin and Prof. Elisa M. Konieczko Explored construction of an electrical pulse stimulation system to exercise skeletal muscle cells in vitro	May 2024 – Oct 2024
Gannon University — BISE VR Lab , VR Research Engineer Supervisor: Prof. Ohu Ikechukwu - Designed gamified safety-awareness training for welding-lathe machines in Unity 3D - Prototyped heat-aware surfaces using Peltier elements to reduce machine operation cost	May 2023 – present

Professional Experience

Lodging Connections LLC , AI Automation Engineer Led a startup team building automation workflows with n8n, Kubernetes, and Make.com	Erie, PA Aug 2025 – present
PanHealth Inc , Head of Engineering Led the US engineering team developing a startup focused on drug delivery for Alzheimer’s patients	Erie, PA June 2025 – present
E3 Company — Erie Technology Incubator , Software Engineer - Led a five-person team building a full-stack web app with JavaScript, MySQL, Redis, and AWS - Built a WebSocket notification and threshold backend, reducing latency by 20%	Erie, PA Dec 2024 – May 2025
PanHealth Inc , Software Engineer Intern - Improved legacy .NET 4 / Web Forms codebases - Developed an MVC pattern for seamless medication viewing and interaction	Remote Dec 2024 – June 2025

Publications

Towards a serverless intelligent firewall: AI-driven security, and zero-trust architectures M. A. R. Chowdhury, Hoang Nam Dang , R. Bazan-Antequera, M. S. Khan, M. R. Karim, S. Manakkadu (2025 IEEE 12th International Conference on Cyber Security and Cloud Computing (CSCloud))	2025
Predictive modeling of student transfer intentions: A comprehensive machine learning approach with novel feature selection methodology Hoang Nam Dang , L. Zhou (Proceedings of the 2nd International Conference on Education and Society (ICES 2026))	Jan 2026

Selected Projects

Cognitive Load Measurement Project Lead · Next.js, Python, Vision Language Models · Sponsored by Lockheed Martin - Developing a bias-aware assessment system using VLMs to analyze candidate micro-expressions and sentiment - Implementing human-in-the-loop validation to meet ethical AI standards for hiring	Jan 2024 – present
Heuristic Encoding Problem in GPT-2 Collaboration with Prof. Gustavo Sandoval · NYU Courant Investigating GPT-2 failures on numeric comparisons (e.g., 9.11 vs 9.8) using mechanistic interpretability, probing, and attention-head analysis	Sept 2025 – present

Volunteer Experience

Cohere Community Lab, Researcher

Remote

- Research on efficient vision-language models with collaborators at Cohere Community Research Lab

Jan 2025 – present

Skills

Programming Languages: Python, C/C++, MATLAB, HTML

Machine Learning: PyTorch, TensorFlow, scikit-learn, Pandas, NumPy